

whether any other device has received the packet during device-to-device communication or not.

A higher-layer DCT protocol can be designed in two different ways. In one, the protocol is dependent on unidirectional packet communication. For example, if B receives a packet sent by A, it is not necessary for A to receive B's packet for the DCT protocol to function properly. Almost all the implemented protocols fall in this category. However, there is a design (called DESIRE) by the PRIVATICS Team from Inria, France, where the mutual exchange of packets is mandatory for detection of 'close contact' events.

Each BLE packet includes a 48-bit MAC-address called the BLE MAC. Interestingly, it is also a pseudo-random number. The BLE protocol layer rotates this address at a regular interval.

Consider a situation where the proximity identifier is rotated by a DCT app after a fixed duration of ten minutes and the BLE MAC remains unchanged at the boundary of the ten-minute interval. Even when the BLE MAC changes at an interval of ten minutes, but the intervals are not aligned with the intervals of proximity identifier (see Table 1), an observer can easily link all the packets and identify that they have been emitted by the same device. This is a type of linkage attack.

This is a potential privacy risk; many currently rolled out automated contact tracing systems suffer from this issue. A typical example is BlueTrace, the protocol underneath Singapore's TraceTogether app, which has not taken any special measure to counter this type of attack. On the other hand, the Apple Google Exposure Notification Framework has taken care of this issue by ensuring that the MAC address changes whenever the proximity identifier is rotated by the cryptographic API-layer. However, the way it has currently been implemented (by restarting the BLE layer every time the proximity identifier changes) appears suboptimal due to the associated performance and latency issues.

An important aspect that every DCT app must consider during design and implementation is to take care of the signal strength variations in BLE communications. The signal strength can be utilised as a measure of distance between the two devices and log entries in one device only when the other device is close enough (say within a distance of two metres). At the same time, we must keep in mind that the devices from different manufacturers use different levels of signal strengths and there is no uniformity or standardisation.

This means every app must be well calibrated to interpret the signal strengths from nearby devices properly - else there will be either too many 'false positives' (where devices are too far apart in reality but the apps still log the encounters) or too many 'false negatives' (where genuine close contact events would be missed due to conservative assumptions on signal strengths). TraceTogether has published extensive data on the signal strength calibration performed before the launch of the app.

Let us investigate the payload part of a BLE packet. There is no uniform agreement about its structure among different Covid-19 apps and hence it varies from one DCT protocol to the other. Undoubtedly, the most important part of the payload is the pseudo-random proximity identifier. Again, the length of the identifier is not the same across different systems. Apart from the identifier, most of the protocols have a provision to include certain meta-data like the transmitting signal strength, protocol version, time-stamp, etc. It is advisable

to encrypt the meta-data so that no active attacker (an adversary who can not only eavesdrop but also modify a packet and resend) may create a spurious packet by combining the identifier from one packet and meta-data from another.

In some protocols the payload also contains an additional Message Authentication Code to ensure the authenticity of the entire payload, irrespective of whether the packet includes any meta-data or not.

Way forward

What can be the future direction of BLE technology when it comes to the use-case of automated contact tracing systems?

The Bluetooth Special Interest Group (Bluetooth SIG) has already announced that they would soon come up with a specification targeted for the wearable devices so that exposure notification functionality can be extended for such usage in embedded systems. We may possibly expect that the current drawbacks like unpredictable rotation frequency of BLE MAC address, lack of native encryption support, or packet-size limitation would get addressed in future versions. There can also be better predictability of signal strength variation over distance for each BLE implementation, irrespective of the OS-type (Android, iOS, or any other), the exact version, or the manufacturer of the device.

While we wait to see these changes, who knows, we may be surprised to see an altogether new technology replacing the Bluetooth Low Energy and giving us a better alternative in the near future! 

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